Jon Wick

Bellingham WA, 98225 | jonwick@wickbioscience.com | wickbioscience.com
github.com/Jon-S-Wick | linkedin.com/in/jonathan-wick-75aa52277 | Orcid ID:0009-0006-6282-2458

PROFESSIONAL SUMMARY

Passionate molecular and computational research scientist with experience in Western blots, PCR, and sterile cell culture, as well as genomic analysis. I started research in high school investigating novel methods for species tracking using eDNA sentinals at RAIN Incubator. Following RAIN experience I participated in education research evaluating learning outcomes in student-led robotics. Most recently I have been investigating RNA interference pathways in *T. thermophila* and protein degradation pathways in *C. elegans*. I am looking to utilize the skills computational and molecular biology skills I have developed to characterize cancers and develop precision theraputics

SKILLS AND TECHNOLOGIES

Wet Lab
- Western Blot
- PCR
- Reverse Transcriptase PCR
- Cell Culture
- Sterile Technique
- Nanodrop

Bioinformatics
- Genomic Data Analysis
- DESeq2
- samtools/bedtools
- Data Visualization (ggplot2, matplotlib)
- RNA-seq

Programming Languages
- Python
- R
- Java
- C/C++
- SQL
- Bash

Software
- Benchling
- CLI/Git/SSH
- ImageJ
- Jupyter
- Microsoft Office
- SnapGene
- Matlab

INDEPENDENT BIOLOGY RESEARCH EXPERIENCE

Undergraduate Researcher – Lee lab, WWU JUN 2024 - PRESENT
Dr. Suzanne Lee and Dr. Dan Pollard

- Investigated roles of small non-coding RNAs in an RNAi pathway maintaining genome integrity in *Tetrahymena thermophila*, combining computational biology with molecular biology techniques.
- Developed custom bioinformatics pipelines in Python and R for genome-wide sRNA biogenesis loci identification and differential expression analysis.
- Maintained live cell cultures under sterile conditions and performed protein analysis via Western Blot to characterize RNAi pathway components.
- Presented findings at lab meetings, journal clubs, and the WWU SURP poster session.
- Collaborated in a team environment to advance the understanding of RNAi pathways.
- Prepared media, buffers, reagents, and performed lab duties for daily lab operations.

Biotech Research Intern, RAIN Incubator – Tacoma, WA MAY 2022 - DEC 2022
Dr. Jennifer McKee-Johnson

- Designed and executed pilot PCR-based experiments for tracking European Green Crab migration.
- Gained foundational experience including hypothesis development, literature review, and experimental planning.

RESEARCH BASED COURSES

Advanced Molecular and Cell w/lab – BIOL 487, WWU SPRING 2025
Dr. Lina Dahlberg

- Investigated ER-associated protein degradation (ERAD) via E2 ubiquitin ligase *ubc-6* in *C. elegans*, a pathway directly relevant to protein quality control in cancer and neurodegeneration.
- Performed Western Blots, Reverse Transcriptase PCR, and endpoint PCR to quantify misfolded protein accumulation in *ubc-6* knockout lines.
- Co-authored and presented a poster for the Northwest Worm Meeting and co-wrote a micropublication on research findings.
- Maintained detailed electronic lab notebooks using Benchling.

Genomic Data Analysis – BIOL 477, WWU WINTER 2025
Dr. Matthew Zinkgraf

- Conducted mRNA differential expression analysis on *Tetrahymena thermophila* datasets using DESeq2, identifying differentially expressed genes across experimental conditions.
- Processed and analyzed genomic data in a Unix environment using Bash, Python, and high-performance computing resources.
- Communicated findings through a written research paper and oral presentation.

EDUCATION

B.S. Molecular and Cellular Biology, Computer Science minor Sept. 2023 - June 2026
Western Washington University – Bellingham, WA

PUBLICATIONS

Understanding the Leadership Structure and Mentoring Model of an Extracurricular Robotics Team: Key Findings from a Case Study March 2024
SIGCSE 2024: Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2
Hitender Oswal, ***Jon Wick***, Seth Tandon, Ashley Brewster, Sushil K Oswal
DOI: 10.1145/3626253.3635603

POSTERS

Re-identification of the genomic origins of small RNAs in the model organism *Tetrahymena thermophila* and characterization of those dependent on genome protective *RDN2*
2025 SURP poster session, October 2025. **Jon Wick** *et al.*

The Role of E2 Ligases ubc-6 and 7 in ERAD in C. elegans
Northwest Worm Meeting, June 2025. **Jon Wick** *et al.*

Remapping previously sequenced 23-24 nucleotide sRNAs to the 2020 Tetrahymena thermophila genome
WWU SURP poster session, September 2024. **Jon Wick** *et al.*

AWARDS

Outstanding Graduate 2026
Western Washington University, Dept of Biology

Declan Barron Memorial Scholarship 2025
Western Washington University, Dept of Biology

WWU Biology Summer Undergraduate Research Program award 2025
Western Washington University, Dept of Biology

ADDITIONAL EXPERIENCE

Club Lead, WWU – Molecular Bioscience Club AUG 2025 - JUN 2026

- Organize events to build community and provide information about programs, research opportunities, and professional development to students of all course standings.
- Assisted students in getting into research labs.
- Collaborated with faculty within and outside of WWU on events.

Computing Education Research – Sushil K. Oswal, UW FEB 2023 - MAY 2024

- Co-authored and published a paper at the ACM SIGCSE conference on mentorship structures in STEM education.
- Performed qualitative analysis on interviews and survey responses.

Programming Lead – SOTAbots FIRST Team 2557 SEP 2019 - JUN 2023

- Developed code used in the semi-finals of the 2022 World Championship of FIRST Robotics Competition and 2023 district championship.
- Taught students to program in Java using industry-standard design patterns.
- Coordinated with other leads on the team to develop a robot, build community, perform local outreach, and obtain funding.